225819: feldspathic sandstone, Silvergrass prospect

(Bellary Formation?, Fortescue Group, Fortescue Basin, Capricorn Orogen)

Location and sampling

Mount Bruce (SF 50-11), McRae (2453)
MGA Zone 50, 587738E 7535155N

Sampled on 13 February 2017

This sample was collected from the 2768.08 – 2768.15 m depth interval of drillcore DD85SGS1, drilled in 1985 by CRA Exploration Pty Ltd at their Silvergrass prospect (Meakins, 1988). The drillhole is located on Hamersley Station, about 5.3 km north-northwest of Four Corners Bore, 4.7 km east-southeast of Kangeenareena Well, and 1.9 km east of Cattle Well.

Geological context

The unit sampled is a feldspathic sandstone within a sedimentary unit that overlies granitic basement rocks of the Pilbara Craton. The unit is overlain by the 2775–2772 Ma Mount Roe Basalt of the Fortescue Group (Arndt et al., 1991; Wingate, 1999; Howard, 2021). The sample was collected near the base of the sedimentary unit, which is 22 m thick and composed of siliciclastic sedimentary rocks. The sandstone is texturally immature and contains angular and rod-shaped quartz grains, some with cuspate margins, and abundant warped mica grains. These grains would be consistent with derivation from basement granitic rocks such as the medium-grained, foliated, chloritized granodiorite at the base of this drillcore (Meakins, 1988) or alternatively could be of volcanic origin. This sandstone was sampled to provide age constraints on the Bellary Formation of the Fortescue Group, although it is possible that the unit is older than the Bellary Formation.

Petrographic description

The sample is an altered feldspathic sandstone containing layers of siltstone (Fig. 1), and consists mainly of sericitic matrix, with <20% micas, <15% quartz, 5–7% feldspar, and minor opaque minerals, pyrite, and zircon. Sandstone layers are 2 to >10 mm thick, and include medium, fine and very fine-grain sizes. Some show gradation in grain size. Lamination within beds is subtle, although contacts between siltstone and sandstone are sharp and planar. Silty layers consist mainly of sericite and tiny opaque grains, together with minor amounts of the detrital minerals present in the sandstone. Quartz grains are angular, transparent, inclusion-poor, non-strained, and range up to 0.4 mm in size. Slightly bent laths of mica, mostly colourless chlorite (probably after biotite), are up to 0.65 mm long and variably aligned parallel to bedding. Angular microcline and less abundant plagioclase grains are up to 0.35 mm in size. Detrital grains are enclosed by almost massive sericite, within which vague grain shapes of unknown origin are discernible. Opaque grains are up to 0.4 mm in size, and were probably originally magnetite but have been replaced by titanite and oxide minerals. Disseminated anhedral grains of pyrite are up to 0.2 mm in size. The sandy layers are moderately sorted although detrital grains are angular. Based on the abundance of mica laths and presence of microcline, the provenance is most likely granitic. The sample is not obviously deformed or metamorphosed.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque and mainly subhedral to euhedral. The crystals are up to 300 µm long, and mainly elongate, with aspect ratios up to 8:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 5–7 November 2020, using SHRIMP-B. Twenty-four analyses of the M257 standard were obtained during the session, of which 20 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.16% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Thirty-two analyses were obtained from 28 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3), and can be divided into three groups, based on their $^{207}Pb^{*}/^{206}Pb^{*}$ ratios.

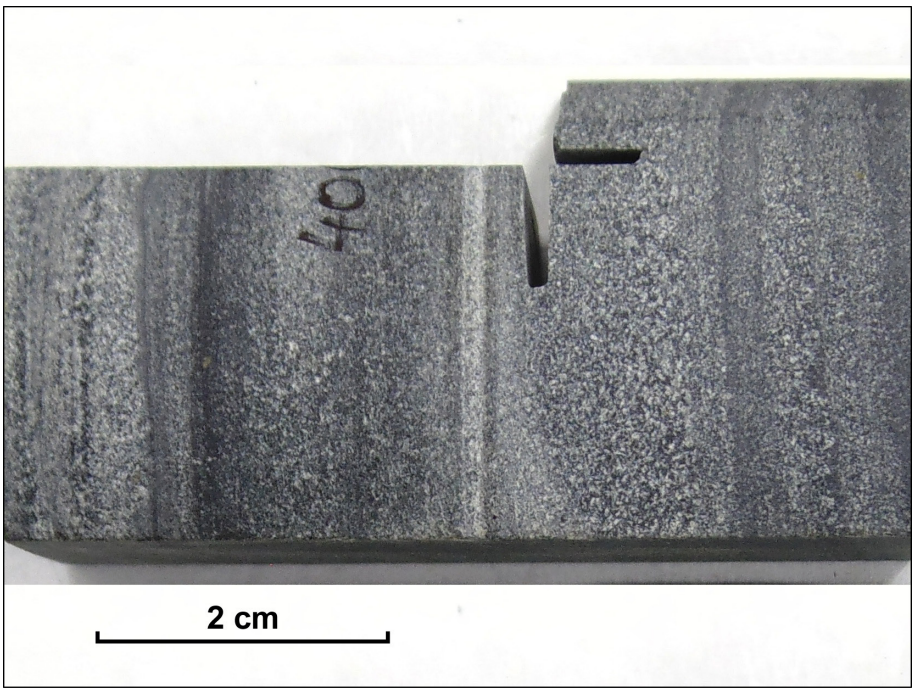

Figure 1. Drillcore image for sample 225819: feldspathic sandstone, Silvergrass prospect

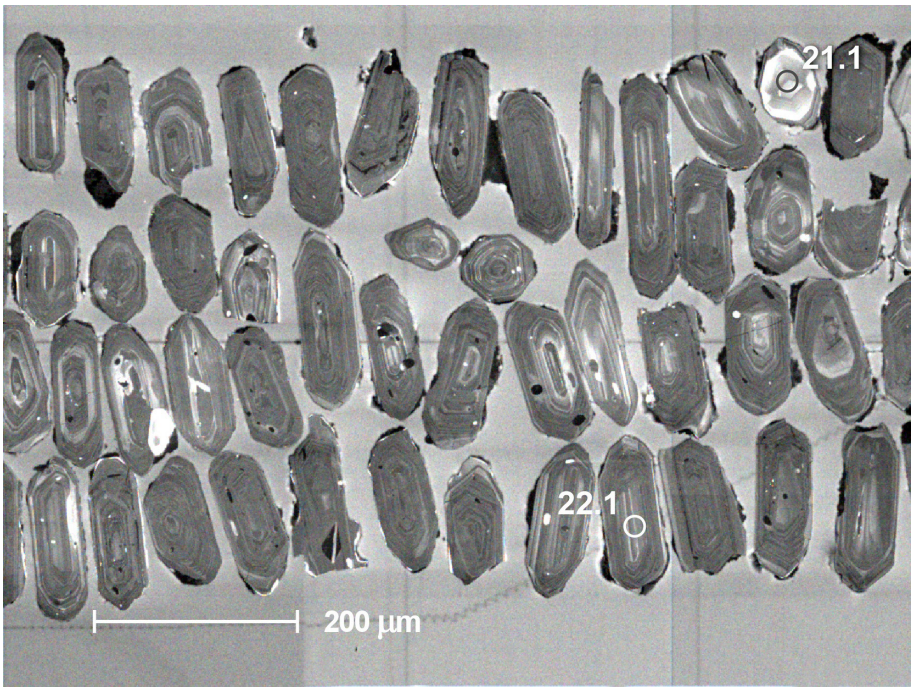

Figure 2. Cathodoluminescence image of representative zircons from sample 225819: feldspathic sandstone, Silvergrass prospect. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 225819: feldspathic sandstone, Silvergrass prospect

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
Y	3	3.1	351	329	0.94	0.052	1.709	0.017	0.22710	0.00091	1.710	0.017	0.22664	0.00093	2968	24	3029	7	2.0
S	19	19.1	130	101	0.77	0.020	1.652	0.023	0.23095	0.00251	1.652	0.023	0.23077	0.00251	3051	34	3057	17	0.2
S	10	10.1	336	311	0.93	0.068	1.641	0.016	0.23172	0.00094	1.642	0.016	0.23111	0.00096	3066	24	3060	7	-0.2
S	15	15.1	210	34	0.16	0.049	1.689	0.019	0.23217	0.00117	1.690	0.019	0.23174	0.00119	2997	28	3064	8	2.2
S	6	6.1	358	35	0.10	-0.014	1.648	0.016	0.23182	0.00091	1.648	0.016	0.23195	0.00092	3058	24	3066	6	0.3
S	7	7.1	112	73	0.65	0.000	1.645	0.024	0.23213	0.00159	1.645	0.024	0.23213	0.00159	3062	36	3067	11	0.2
S	13	13.1	351	387	1.10	0.029	1.625	0.016	0.23242	0.00091	1.625	0.016	0.23216	0.00092	3091	25	3067	6	-0.8
S	23	23.1	215	34	0.16	0.131	1.647	0.019	0.23356	0.00117	1.649	0.019	0.23240	0.00122	3056	28	3069	8	0.4
S	12	12.1	419	525	1.25	0.036	1.639	0.015	0.23280	0.00084	1.640	0.015	0.23248	0.00085	3069	23	3069	6	0.0
S	29	18.2	496	264	0.53	0.006	1.649	0.015	0.23268	0.00082	1.649	0.015	0.23263	0.00082	3056	22	3070	6	0.5
S	27	27.1	135	145	1.07	0.038	1.656	0.023	0.23348	0.00146	1.657	0.023	0.23315	0.00148	3044	33	3074	10	1.0
S	16	16.1	120	31	0.26	0.196	1.639	0.023	0.23501	0.00157	1.643	0.024	0.23328	0.00168	3065	35	3075	11	0.3
S	5	5.1	246	32	0.13	0.081	1.685	0.018	0.23405	0.00108	1.686	0.018	0.23334	0.00111	3002	26	3075	8	2.4
S	11	11.1	163	151	0.93	0.121	1.663	0.034	0.23496	0.00131	1.665	0.034	0.23390	0.00137	3032	50	3079	9	1.5
S	26	26.1	238	177	0.75	0.011	1.666	0.018	0.23413	0.00110	1.666	0.018	0.23403	0.00111	3031	27	3080	8	1.6
S	21	21.1	83	58	0.69	-0.094	1.693	0.028	0.23328	0.00187	1.692	0.028	0.23411	0.00193	2994	40	3080	13	2.8
S	24	24.1	210	80	0.38	0.097	1.702	0.020	0.23498	0.00200	1.703	0.020	0.23413	0.00202	2978	28	3081	14	3.3
S	2	2.1	152	253	1.66	-0.054	1.756	0.021	0.23380	0.00141	1.755	0.021	0.23428	0.00144	2907	29	3082	10	5.7
S	22	22.1	187	206	1.11	0.014	1.680	0.020	0.23454	0.00128	1.680	0.020	0.23442	0.00129	3011	29	3083	9	2.3
S	25	25.1	234	176	0.75	0.023	1.658	0.019	0.23474	0.00114	1.658	0.019	0.23454	0.00115	3043	27	3083	8	1.3
S	31	15.2	23	12	0.51	-0.353	1.657	0.051	0.23168	0.00358	1.652	0.051	0.23480	0.00400	3052	75	3085	27	1.1
S	8	8.1	527	173	0.33	0.043	1.676	0.015	0.23520	0.00077	1.677	0.015	0.23481	0.00078	3015	22	3085	5	2.3
S	14	14.1	160	163	1.02	0.000	1.699	0.022	0.23503	0.00134	1.699	0.022	0.23503	0.00134	2984	30	3087	9	3.3
S	9	9.1	134	82	0.62	0.035	1.655	0.039	0.23571	0.00142	1.656	0.039	0.23540	0.00144	3046	57	3089	10	1.4
S	1	1.1	81	74	0.92	0.000	1.681	0.028	0.23645	0.00189	1.681	0.028	0.23645	0.00189	3009	41	3096	13	2.8
S	4	4.1	119	22	0.18	0.000	1.701	0.025	0.23720	0.00161	1.701	0.025	0.23720	0.00161	2981	35	3101	11	3.9
S	32	28.1	1081	317	0.29	0.003	1.655	0.019	0.23901	0.00194	1.655	0.019	0.23899	0.00194	3047	28	3113	13	2.1
S	20	20.1	107	82	0.76	-0.045	1.588	0.024	0.25358	0.00334	1.587	0.024	0.25397	0.00335	3150	37	3210	21	1.9
S	17	17.1	62	19	0.31	0.000	1.449	0.027	0.29124	0.00261	1.449	0.027	0.29124	0.00261	3383	49	3424	14	1.2
P	28	3.2	760	390	0.51	0.010	1.755	0.014	0.20655	0.00058	1.755	0.014	0.20646	0.00058	2907	19	2878	5	-1.0
P	30	18.3	80	64	0.81	0.291	1.727	0.035	0.23036	0.00229	1.732	0.035	0.22779	0.00253	2938	48	3037	18	3.2
P	18	18.1	209	51	0.24	0.050	1.635	0.030	0.23053	0.00117	1.636	0.030	0.23009	0.00119	3075	45	3053	8	-0.7

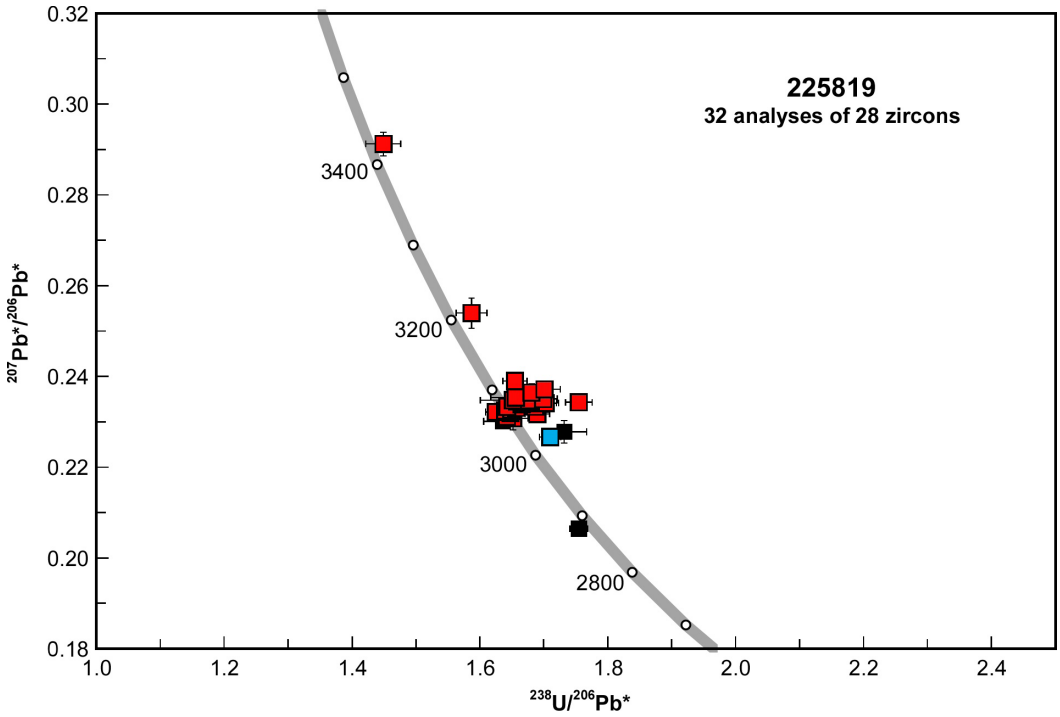

Figure 3. U–Pb analytical data for zircons from sample 225819: feldspathic sandstone, Silvergrass prospect. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); black squares indicate Group P (radiogenic-Pb loss)

Group Y comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 3029 ± 7 Ma (1σ).

Group S comprises 28 analyses of 27 zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 3424–3057 Ma.

Group P comprises three analyses of two zircons (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 3053–2878 Ma.

It is possible that all of the analyses in Groups Y and S are of unmodified detrital zircons, in which case the date of 3029 ± 7 (1σ) for the single analysis in Group Y represents a maximum depositional age. A more conservative estimate of the maximum depositional age can be based on the weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 3075 ± 4 Ma (MSWD = 1.4) for the youngest 25 analyses in Group S. This date is interpreted as the age of the dominant zircon-crystallizing rock in the detrital source region, which is probably the underlying granodiorite, as suggested by Meakins (1988).

The dates of 3053–2878 Ma for three analyses in Group P are significantly younger than additional analyses of the same zircons, and are interpreted to reflect loss of radiogenic Pb.

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Howard, HM 2021, 225819: feldspathic sandstone, Silvergrass prospect; Geochronology Record 1815: Geological Survey of Western Australia, 4p.

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